Семинар 4 27.09.2024

Задача 1

$y_t = 4 + y_{t-1} - 0{,}4y_{t-2} + \varepsilon_t$ ← $WN \sim (0;\ \sigma_\varepsilon^2 = 1)$

$y_t - y_{t-1} + 0{,}4y_{t-2} = 4 + \varepsilon_t$

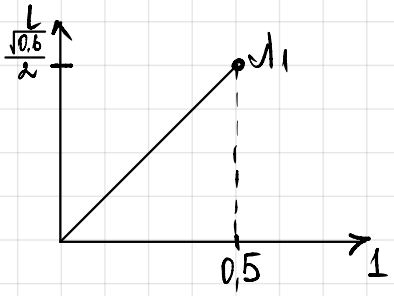
$\lambda^2 - \lambda^1 + 0{,}4\lambda^0 = 0$

$t - 2024 = n$
$t = n + 2024$
$t - 1 = n + 2023$

$y_t = y_{t-1} + y_{t-2024}$
$\lambda^{2024} - \lambda^{2023} - \lambda^0 = 0$
$y_{n+2024} - y_{n+2023} - y_n = 0$

$D = b^2 - 4ac = 1 - 4 \cdot 0{,}4 = -0{,}6$

$\lambda_{1,2} = \dfrac{-b \pm \sqrt{D}}{2a} = \dfrac{1 \pm \sqrt{-0{,}6}}{2} = 0{,}5 \pm i \cdot \dfrac{\sqrt{0{,}6}}{2}$



$|\lambda_{1,2}| = \sqrt{0{,}5 + \left(\pm\dfrac{\sqrt{0{,}6}}{2}\right)^2} = \sqrt{0{,}5^2 + \dfrac{0{,}6}{4}} = \sqrt{0{,}4} < 1$

⇓ стационарен

1) $\mathbb{E}\, y_t = 4 + \mathbb{E}\, y_t - 0{,}4\mathbb{E}\, y_t = 0$

$0{,}4\,\mathbb{E}\, y_t = 4$

$\mathbb{E}\, y_t = 10$

$cov(\varepsilon_t; \varepsilon_t) = \sigma_\varepsilon^2$
$cov(\varepsilon_t; y_t) = \mathbb{E}\varepsilon_t\,\mathbb{E}\, y_t$ (=0)

2) У-У: y_t $\quad \mathbb{E}\, y_t \cdot y_t = \mathbb{E}\, 4 \cdot y_t + \mathbb{E} y_{t-1} \cdot y_t - 0{,}4\mathbb{E}_{t-2} \cdot y_t + \mathbb{E}\varepsilon_t \cdot y_t$

Опускаем константу

y_{t-1} $\quad \begin{cases} \gamma_0 = \gamma_1 - 0{,}4\gamma_2 + 1 \\ \gamma_1 = \gamma_0 - 0{,}4\gamma_1 \\ \gamma_2 = \gamma_1 - 0{,}4\gamma_0 \end{cases} \quad \begin{cases} 1{,}4\gamma_1 = \gamma_1 - 0{,}4 \cdot 0{,}4\gamma_1 + 1 \\ \gamma_0 = 1{,}4\gamma_1 \\ \gamma_2 = \gamma_1 - 0{,}4 \cdot 1{,}4\gamma_1 = 0{,}44\gamma_1 \end{cases}$

y_{t-2}

АКов: $\begin{cases} \gamma_1 = 1{,}74 \\ \gamma_0 = 2{,}43 \\ \gamma_2 = 0{,}76 \end{cases} = \Delta y_t$

АКорФ:

$\rho_1 = \dfrac{\gamma_1}{\gamma_0} = 0{,}7$

$\rho_2 = \dfrac{\gamma_2}{\gamma_0} = 0{,}3$

PACF(1): $y_t = \beta_1 y_{t-1} + u_t$

y_{t-1}

$\gamma_1 = \beta_1 \cdot \gamma_0 \qquad \beta_1 = \frac{\gamma_1}{\gamma_0} = 0{,}7$

PACF(2): $y_t = \alpha_1 \cdot y_{t-1} + \alpha_2 y_{t-2} + u_t$ → не коррелирует с y_t

(PACF(3): $y_t = \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \varphi_3 y_{t-3} + u_t$)

y_{t-1}: $\gamma_1 = \alpha_1 \gamma_0 + \alpha_2 \gamma_1$

y_{t-2}: $\gamma_2 = \alpha_1 \gamma_1 + \alpha_2 \gamma_0$

$\Rightarrow$ $\alpha_1 =$ число 1, $\alpha_2 =$ число 2 = PACF

Задача 2

$y_t = 2 + 0{,}4 y_{t-1} - 0{,}03 y_{t-2} + \varepsilon_t + \varepsilon_{t-1}$ $\Rightarrow$ ARMA(2;1)

$y_t (1 - 0{,}4L + 0{,}03L^2) = 2 + \varepsilon_t (1 + L)$

$y_t (1 - 0{,}1L)(1 - 0{,}3L) = 2 + \varepsilon_t (1 + L)$

$y_t = \frac{2}{(1-0{,}1L)(1-0{,}3L)} + \frac{\varepsilon_t (1+L)}{(1-0{,}1L)(1-0{,}3L)}$

$y_t = 2 \cdot (1 + 0{,}1L + (0{,}1L)^2 + (0{,}1L)^3 + \ldots) \cdot (1 + 0{,}3L + (0{,}3L)^2 + (0{,}3L)^3 + \ldots) +$
$+ \varepsilon_t \cdot (1 + L) \cdot (1 + 0{,}1L + (0{,}1L)^2 + (0{,}1L)^3 + \ldots) \cdot (1 + 0{,}3L + (0{,}3L)^2 + (0{,}3L)^3 + \ldots)$

$y_t = 2 \cdot (1 + 0{,}1 + 0{,}1^2 + 0{,}1^3 + \ldots) \cdot (1 + 0{,}3 + 0{,}3^2 + 0{,}3^3 + \ldots) +$
$+ \varepsilon_t (1 + 0{,}3L + 0{,}1L + L + 1 \cdot 0{,}1L \cdot 0{,}3L + 0{,}3L^2 + 0{,}1L^2 + 0{,}01L^2 + 0{,}09L^2 + \ldots)$

$y_t = \frac{2}{0{,}9 \cdot 0{,}7} + \varepsilon_t + 1{,}4 \varepsilon_{t-1} + 0{,}53 \varepsilon_{t-2} + \text{число}_1 \cdot \varepsilon_{t-3} + \text{число}_2 \cdot \varepsilon_{t-4} + \ldots$